

Research Interests

My research interests lie in group (co)homology of congruence subgroups of arithmetic groups and virtual duality groups. I am also interested in questions related to homological stability for families of groups.

Education

2021-Present **Ph.D in Mathematics**, *University of Oklahoma*

Advisor: Peter Patzt

2019-2021 **M.S. in Mathematics**, *American University of Beirut*

Advisor: Hazar Abu-Khuzam

Thesis Title: Structure of certain rings with conditions on nonperiodic elements and certain Von-Neumann II-regular rings with prime centers

2015-2018 **B.S. in Mathematics**, *Lebanese University*

Research Papers

1. **Steinberg homology of $\Gamma_{0,n}^{\pm}(p)$ as a Hopf module and Hecke congruence subgroups** (in preparation), with Peter Patzt
2. **Codimensions one and two cohomology of Hecke congruence subgroups** (under review), with Jeremy Miller, arXiv:2606.23519
3. **Top-dimensional rational cohomology of the congruence subgroup $\Gamma_{0,n}^{+}(p)$** (under review), arXiv:2605.23526

Honors, Awards, and Fellowships

University of Oklahoma:

- 2026
 - MGSA Good Mentor Award
 - Thomas J. Hill Graduate Fellowship in Mathematics
 - T. Jay Thompson Scholarship Award
 - Armstrong Memorial Scholarship Award
 - SHS-OG&E Energy Corp Scholarship Award
 - Dodge Family Completion Fellowship Award
- 2025
 - David and Judi Proctor Scholarship
 - Robert Lee Loshbough Endowed Scholarship Fund
 - Arts and Sciences - 31171
 - Dodge Family College of Arts and Sciences Travel Assistance Program

- 2024 ○ **Summer Graduate Fellowship**
○ **Dodge Family College of Arts and Sciences Travel Assistance Program**
- 2023 ○ **Conrad and Ellen Masterson Fellowship in Mathematics**
○ **Dodge Family College of Arts and Sciences Travel Assistance Program**
- 2022 **C. Eugene Springer Mathematics Scholarship**
- 2021-2025 **Fellowship from the Mathematics department**
- 2021-Present **Graduate Teaching Assistantship**

American University of Beirut:

- 2029-2021 **Graduate Assistantship**

Research Talks and Poster Presentations

- Aug. 2026 **Cohomology of Hecke congruence subgroups** (lightning talk), *Summer 2026 Wasatch Topology Conference, Utah*
- Jun. 2026 **On the rational (co)homology of Hecke congruence subgroup**, *Topology Students Workshop, Vanderbilt University*
- May 2026 **Cohomology of congruence subgroup $\Gamma_{0,n}(p)$** , *International Seminar on Quantization, Geometry, and Dynamics, University of Texas Rio Grande Valley (virtual)*
- Apr. 2026 **Top-dimensional cohomology of congruence subgroup $\Gamma_{0,n}(p)$** , *Topology seminar, Purdue University*
- Feb. 2026 **Top-dimensional cohomology of congruence subgroup $\Gamma_{0,n}(p)$** , *Topology seminar, University of Michigan*
- Oct. 2025 **Top-dimensional rational cohomology of the congruence subgroup $\Gamma_{0,n}(p)$** , *Geometry and Topology Seminar, University of Oklahoma*
- Oct. 2025 **Top-dimensional rational cohomology of the congruence subgroup $\Gamma_{0,n}(p)$** (poster presentation), *Perspectives on the cohomology of general linear groups, University of Münster*
- Jul. 2025 **Double Steinberg Coinvariants**, *Scissors Congruence and K-theory Conference, University of Pennsylvania*

Expository Talks

- Nov. 2025 **Borel–Serre duality**, *Student Topology Seminar, University of Oklahoma*
- Nov. 2023 **High Dimensional Cohomology of Arithmetic Groups**, *The 2023 Binghamton University Graduate Combinatorics, Algebra, and Topology Conference, Binghamton University*
- Dec. 2022 **Introduction to Spectral Sequences**, *Student Presentation Seminar, University of Oklahoma*

University of Oklahoma:

- 2025-2026 **Treasurer of the University of Oklahoma AWM Chapter**
- Spring 2025 **Mentor, Directed Reading Program**, *On the Proof of Abel and Ruffini*

- 2024–2025
 - Graduate Student Senate
 - Outreach of the University of Oklahoma AWM Chapter
 - Co-organizer of the University of Oklahoma Graduate Math Fest (an annual Graduate Student Conference at the University of Oklahoma)
- 2023–2024
 - Co-founder & President of the University of Oklahoma AWM Chapter
 - Co-founder of the AWM Mentorship Program for undergraduate students
 - Secretary of the Math Graduate Student Association
- 2022–2023 Member of the Graduate Advocacy Committee

Teaching Experience

Instructor of Record:

- SP26 Differential & Integral Calculus II (HONORS), *University of Oklahoma*
- FA25 Functions and Modeling for Business, Life and Social Sciences. (large section ~ 80 students), *University of Oklahoma*
- SP24 Calculus I for Business, Life and Social Sciences, *University of Oklahoma*
- FA23 Functions and Modeling for Business, Life and Social Sciences., *University of Oklahoma*
- SU23 Calculus & Analytics Geometry IV, *University of Oklahoma*
- SP23 Functions and Modeling for Business, Life and Social Sciences., *University of Oklahoma*
- SU22 Calculus & Analytics Geometry I, *University of Oklahoma*

Discussion Leader:

- SP25 Differential & Integral Calculus II (HONORS), *University of Oklahoma*
- FA24 Differential & Integral Calculus II (HONORS), *University of Oklahoma*
- FA22 Differential & Integral Calculus I, *University of Oklahoma*
- SP22 Calculus and Analytics Geometry I, *University of Oklahoma*

Grader:

- SP25 Topology II, *University of Oklahoma*
- FA24 Topology I, *University of Oklahoma*
- FA22 Introduction to Analysis I, *University of Oklahoma*
- FA21 Calculus & Analytics Geometry IV, *University of Oklahoma*
- FA21 Differential & Integral Calculus III (HONORS), *University of Oklahoma*
- FA20 Rings & Modules I, *American University of Beirut*
- FA19, SP20 Introduction to Analysis I, *American University of Beirut*

Service and Outreach

University of Oklahoma:

- 2025–2026 Treasurer of the University of Oklahoma AWM Chapter

- Spring 2025 **Mentor, Directed Reading Program, On the Proof of Abel and Ruffini**
- 2024–2025
- **Graduate Student Senate**
 - **Outreach of the University of Oklahoma AWM Chapter**
 - **Co-organizer of the University of Oklahoma Graduate Math Fest** (an annual Graduate Student Conference at the University of Oklahoma)
- 2023–2024
- **Co-founder & President of the University of Oklahoma AWM Chapter**
 - **Co-founder of the AWM Mentorship Program for undergraduate students**
 - **Secretary of the Math Graduate Student Association**
- 2022–2023 **Member of the Graduate Advocacy Committee**

Conferences and Workshops attended

- May. 2026 **Configuration spaces summer school**, *University of Michigan*
- Jun. 2025 **Polylogarithms, homology of linear groups, and Steinberg modules**, *SwissMAP Research, Station in Les Diablerets*
- May. 2025 **Summer school: Homological aspects of groups and their classifying spaces**, *SwissMAP Research, Station in Les Diablerets*
- Feb. 2025 **Redbud Topology Conference Spring 2025**, *University of Arkansas*
- Jul. 2024 **Collaborative Workshop in K-theory and Scissors Congruence**, *Vanderbilt University*
- Oct. 2023 **The South Central Topology Conference II**, *New Mexico State University*
- Oct. 2023 **Redbud Topology Conference Fall 2023**, *University of Arkansas*
- Jul. 2023 **Stability in Topology, Arithmetic, and Representation Theory 2023**, *Purdue University*
- Jun. 2023 **Master class : Homology of groups and functors**, *Université de Lille*
- April 2023 **Redbud Topology Conference Spring 2023**, *University of Oklahoma*
- Feb. 2023 **The South Central Topology Conference II**, *Texas State University*
- Nov. 2022 **The 2022 Binghamton University Graduate Combinatorics, Algebra, and Topology Conference**, *Binghamton University*
- Oct. 2022 **Redbud Topology Conference Fall 2022**, *University of Arkansas*
- April 2022 **Texas-Oklahoma Representations and Automorphic forms XI**, *Oklahoma State University*